

D-03-005 — Audit access

Only Sales Manager, Comptroller, and Installation Manager receive audit-log access through the application.

D-03-006 — Denied-action logging

Permission failures and restricted-access attempts are logged even when the requested operation is denied.

D-03-007 — Audit access auditing

Viewing, searching, exporting, verifying, and retrieving audit records are themselves audited.

D-03-008 — Fail-closed behavior

A required business operation must not succeed if its required audit event cannot be created, except through a separately approved emergency procedure.

D-03-009 — Sensitive-data minimization

Audit payloads must contain enough information to reconstruct activity without storing unnecessary secrets or sensitive values.

D-03-010 — Historical immutability

Audit records cannot be repaired by editing historical rows. Corrections are recorded as new events.

17. Sprint 03 Acceptance Criteria

Sprint 03 is accepted when:

Audit records are stored separately from ordinary business records.

Audit events are append-only.

Ordinary users cannot edit or delete audit events.

Every event has a unique sequence number.

Every event has a unique event identifier.

Every event references the previous event hash except the genesis event.

Event hashes use SHA-256.

Canonical event serialization is defined and deterministic.

A changed event causes chain verification failure.

A deleted event causes chain verification failure.

An inserted event causes chain verification failure.

A reordered event causes chain verification failure.

Duplicate sequence numbers cause verification failure.

Missing sequence numbers cause verification failure.

Successful logins are logged.

Failed login attempts are logged.

Logout is logged.

Views are logged.

Searches are logged.

Creates are logged.

Edits are logged.

Approvals are logged.

Exports are logged.

Prints are logged.

Downloads are logged.

Emails are logged.

Signatures are logged.

Permission failures are logged.
Restricted-access attempts are logged.
Administrative changes are logged.
Audit-log access is restricted by role.
Audit-log searches are permission-controlled.
Audit-log exports are permission-controlled.
Audit-log verification results are preserved.
Audit records are permanently retained.
Archive segments include integrity metadata.
Archived records can be restored and verified.
Audit failures create administrative exceptions.
Required business operations fail closed if required audit logging fails.
Audit payloads do not contain passwords, session tokens, or private authentication secrets.
Audit events reference the relevant project, customer, document, payment, report, or financial period where applicable.

18. Sprint 03 Test Cases

Hash-chain tests

Create a genesis event.
Append a valid event.
Verify the complete chain.
Modify an event and confirm verification failure.
Remove an event and confirm verification failure.
Insert an event and confirm verification failure.
Reorder events and confirm verification failure.
Duplicate a sequence number and confirm verification failure.
Use an invalid previous-event hash and confirm verification failure.

Access-event tests

Successful login creates an audit event.
Failed login creates an audit event.
Logout creates an audit event.
Authorized record view creates an audit event.
Unauthorized record view creates a denied-access event.
Search creates an audit event.
Export creates an audit event.
Print creates an audit event.
Download creates an audit event.

Business-event tests

Customer creation is logged.
Project creation is logged.
Master Sales Record approval is logged.
Master Sales Record reopening is logged.
Version creation is logged.
Document generation is logged.
Signature completion is logged.
Payment verification is logged.
Adjustment approval is logged.
Commission-report approval is logged.
Financial-period locking is logged.

Protection tests

Ordinary application user cannot update an audit event.
Ordinary application user cannot delete an audit event.
Unauthorized role cannot search audit records.
Unauthorized role cannot export audit records.

Database Administrator cannot use ordinary application functions to alter audit history.

Audit-log access is itself audited.

Failed audit insertion prevents the required business action from succeeding.

Retention tests

Audit records remain available after business-record retirement.

Archived records retain their sequence range.

Archive checksums verify successfully.

Archive retrieval creates an audit event.

Chain verification works across hot and archived segments.

19. Sprint 03 Open Questions

These implementation questions do not change approved requirements:

Will the audit log reside in a separate PostgreSQL schema or a separate database?

What exact archive-storage technology will provide immutable retention locking?

What hot-storage period will be used before archival?

What audit verification schedule will be used?

What notification mechanism will report integrity failures?

What emergency audit mode, if any, will be permitted?

What infrastructure-level controls will govern PostgreSQL superuser access?

What exact export formats are required for formal compliance reviews?

What maximum payload size is permitted for an audit event?

What operational process will approve archive restoration?

What recovery-point and recovery-time objectives apply to audit data?

20. Sprint 04 Handoff

Sprint 04 will implement customers, contacts, locations, projects, duplicate detection, project-code generation, and search.

Sprint 04 must create audit events for:

Customer creation

Customer edits

Contact creation and edits

Location creation and edits

Project creation

Project-code reservation

Duplicate-location warnings

Duplicate-location overrides

Project assignment changes

Project cancellation

Project searches

Restricted project-access attempts

Sprint 04 must preserve all audit-event references defined in this bundle.

21. Historical Record

Supersedes

None.

Preserves

GECC-00-project-baseline.md

GECC-01-data-model.md

GECC-02-auth-and-permissions.md

Sprint 00 decisions

Sprint 01 decisions

Sprint 02 decisions

Approved role and permission restrictions
Permanent audit-retention requirement
Changes introduced by this bundle
Defines the Sprint 03 append-only audit-event model.
Defines SHA-256 hash chaining.
Defines canonical event serialization.
Defines audit-event categories and required payloads.
Defines audit search and export behavior.
Defines permanent retention and immutable archival.
Defines integrity verification and failure handling.
Defines Sprint 04 audit integration requirements.
===== END FILE: GECC-03-Audit-Log.md =====
=== BEGIN FILE: GECC-04-Customers-And-Projects.md ===
GECC-04-customers-and-projects.md

GECC Sales Back Office System — Sprint 04 Customers, Contacts, Locations, and Projects

Bundle Metadata

- Bundle name: GECC-04-customers-and-projects.md
- Project: GECC Sales Back Office System
- Company: Go Ecco Climate Control
- Company abbreviation: GECC
- Bundle type: Sprint 04 design and implementation bundle
- Status: Approved design; ready for implementation
- Created date: 2026-09-20
- Previous bundle: GECC-03-audit-log.md
- Next planned bundle: GECC-05-master-sales-record.md
- Database: PostgreSQL
- Primary-key strategy: Numeric BIGINT identity keys
- Business timezone: America/New_York
- System of record: PostgreSQL relational database
- Audit model: Append-only, hash-chained, permanently retained

1. Sprint Objective

Implement customer, contact, project-location, project, assignment, duplicate-detection, project-code, and authorized-search functionality.

Sprint 04 must establish the foundational records used by all later project, document, payment, commission, and financial processes.

2. Sprint 04 Deliverables

Sprint 04 must produce:

1. Customer records
2. Multiple customer contacts
3. Customer contact preferences

4. Project locations
5. State-based location data
6. Customer and location duplicate detection
7. Project records
8. Project assignments
9. Project-status foundation
10. Project-code generation
11. Project-code reservation
12. Cancelled-code protection
13. Active-location duplicate warning and override
14. Authorized search
15. Search filters
16. Role-scoped search results
17. Customer and project access controls
18. Customer and project audit events
19. Database acceptance tests
20. Sprint 05 Master Sales Record handoff

3. Requirements Carried Forward

The implementation must preserve these approved requirements:

- A customer may have multiple projects.
- A project has one customer.
- A project has one project location.
- A project location has separate street, city, state, and ZIP fields.
- The project location's state determines default license artwork.
- Users enter Sales Associate initials.
- The system automatically generates the sequential project number.
- The sequence begins at 101.
- The project-code format is:

```text

Sequential Number-Sales Associate Initials-YYYY-MM-DD

The date is the Master Sales Record creation date.

Project codes must be unique.

Project-code numbers are reserved during creation.

Cancelled project codes cannot be reused.

Revisions preserve the original project code.

The system warns about an existing active project at the same customer location.

A second active project at the same location requires an explanation or correction.

Search results must respect role-based access restrictions.

All access and search activity must be audited.

The relational database is the system of record.

Customer and project records are never silently overwritten.

#### 4. Customer Data Model

##### 4.1 customer

| Field           | Type         | Requirement      |
|-----------------|--------------|------------------|
| id              | BIGINT       | Primary key      |
| customer_number | VARCHAR(50)  | Required, unique |
| customer_name   | VARCHAR(250) | Required         |

|                           |              |                      |
|---------------------------|--------------|----------------------|
| customer_status           | VARCHAR(30)  | Required             |
| customer_folder_reference | TEXT         | Optional             |
| duplicate_check_key       | VARCHAR(500) | Required, indexed    |
| notes                     | TEXT         | Optional             |
| created_at                | TIMESTAMPTZ  | Required             |
| created_by                | BIGINT       | Required foreign key |
| updated_at                | TIMESTAMPTZ  | Required             |
| updated_by                | BIGINT       | Required foreign key |
| is_active                 | BOOLEAN      | Required             |
| retired_at                | TIMESTAMPTZ  | Optional             |
| retired_by                | BIGINT       | Optional foreign key |
| retirement_reason         | TEXT         | Optional             |

The customer number is system-generated and is separate from the project identification code.

#### 4.2 customer\_phone

| Field                   | Type        | Requirement               |
|-------------------------|-------------|---------------------------|
| id                      | BIGINT      | Primary key               |
| customer_id             | BIGINT      | Required foreign key      |
| phone_number            | VARCHAR(50) | Required                  |
| normalized_phone_number | VARCHAR(30) | Required, indexed         |
| phone_type              | VARCHAR(30) | Mobile, home, work, other |
| is_primary              | BOOLEAN     | Required                  |
| is_active               | BOOLEAN     | Required                  |
| Standard audit columns  | —           | Required                  |

#### 4.3 customer\_email

| Field                    | Type         | Requirement           |
|--------------------------|--------------|-----------------------|
| id                       | BIGINT       | Primary key           |
| customer_id              | BIGINT       | Required foreign key  |
| email_address            | VARCHAR(320) | Required              |
| normalized_email_address | VARCHAR(320) | Required, indexed     |
| email_type               | VARCHAR(30)  | Personal, work, other |
| is_primary               | BOOLEAN      | Required              |
| is_active                | BOOLEAN      | Required              |
| Standard audit columns   | —            | Required              |

#### 4.4 customer\_contact

| Field                    | Type         | Requirement          |
|--------------------------|--------------|----------------------|
| id                       | BIGINT       | Primary key          |
| customer_id              | BIGINT       | Required foreign key |
| contact_name             | VARCHAR(200) | Required             |
| contact_type             | VARCHAR(50)  | Required             |
| phone_number             | VARCHAR(50)  | Optional             |
| email_address            | VARCHAR(320) | Optional             |
| is_primary               | BOOLEAN      | Required             |
| preferred_contact_method | VARCHAR(30)  | Optional             |
| contact_status           | VARCHAR(30)  | Required             |
| Standard audit columns   | —            | Required             |

A customer may have multiple contacts.

Only one active primary contact may exist for a given customer and contact type unless an approved exception is recorded.

### 5. Project-Location Data Model

#### 5.1 project\_location

| Field                     | Type         | Requirement          |
|---------------------------|--------------|----------------------|
| id                        | BIGINT       | Primary key          |
| street_address            | VARCHAR(250) | Required             |
| address_line_2            | VARCHAR(250) | Optional             |
| city                      | VARCHAR(100) | Required             |
| state_code                | CHAR(2)      | Required             |
| postal_code               | VARCHAR(20)  | Required             |
| country_code              | CHAR(2)      | Required, default US |
| normalized_street_address | VARCHAR(250) | Required             |
| normalized_city           | VARCHAR(100) | Required             |
| normalized_postal_code    | VARCHAR(20)  | Required             |
| normalized_location_key   | VARCHAR(500) | Required, indexed    |
| Standard audit columns    | —            | Required             |

The state code must use a controlled list of approved state values.

## 5.2 Location normalization

The system must normalize locations for duplicate detection while preserving the original entered values.

Normalization may include:

- Uppercase comparison values
- Standardized street suffixes
- Consistent whitespace
- Consistent punctuation
- Standardized ZIP-code representation
- Separate handling of apartment, unit, lot, or suite information

The original address remains the display and historical value.

## 5.3 Location matching

The system must support:

- Exact normalized-location matching
- Same-customer active-project matching
- Existing active-project matching across customers
- Possible-match warnings for incomplete or similar addresses
- Authorized override with an explanation

The precise fuzzy-matching algorithm remains an implementation decision.

## 6. Project Data Model

### 6.1 project

| Field                       | Type        | Requirement              |
|-----------------------------|-------------|--------------------------|
| id                          | BIGINT      | Primary key              |
| customer_id                 | BIGINT      | Required foreign key     |
| project_code                | VARCHAR(80) | Required, unique         |
| project_location_id         | BIGINT      | Required foreign key     |
| assigned_sales_associate_id | BIGINT      | Required foreign key     |
| current_msr_version_id      | BIGINT      | Optional until Sprint 05 |
| project_status              | VARCHAR(50) | Required                 |
| cancellation_reason         | TEXT        | Required when cancelled  |
| completion_date             | DATE        | Optional                 |
| last_activity_at            | TIMESTAMPZ  | Required                 |
| Standard audit columns      | —           | Required                 |

## 6.2 Initial project statuses

Sprint 04 must support at least:

text

draft  
active  
duplicate\\_review  
cancelled  
superseded  
completed

Later sprints will add controlled lifecycle states for:

Approval  
Contract signing  
Invoice signing  
Installation  
Certificate signing  
Payment  
Commission eligibility  
Final completion  
Sprint 04 must not bypass the future controlled lifecycle.

## 6.3 project\_assignment

| Field                  | Type        | Requirement                        |
|------------------------|-------------|------------------------------------|
| id                     | BIGINT      | Primary key                        |
| project_id             | BIGINT      | Required foreign key               |
| user_id                | BIGINT      | Required foreign key               |
| assignment_type        | VARCHAR(40) | Sales Associate, Technician, other |
| effective_from         | DATE        | Required                           |
| effective_to           | DATE        | Optional                           |
| assignment_status      | VARCHAR(30) | Required                           |
| assigned_by            | BIGINT      | Required foreign key               |
| assignment_reason      | TEXT        | Optional                           |
| Standard audit columns | —           | Required                           |

A project may have multiple technicians.

A project must have one active assigned Sales Associate unless the project is in a draft or exception state.

## 6.4 project\_status\_history

| Field           | Type        | Requirement              |
|-----------------|-------------|--------------------------|
| id              | BIGINT      | Primary key              |
| project_id      | BIGINT      | Required foreign key     |
| previous_status | VARCHAR(50) | Optional                 |
| new_status      | VARCHAR(50) | Required                 |
| changed_at      | TIMESTAMPZ  | Required                 |
| changed_by      | BIGINT      | Required foreign key     |
| change_reason   | TEXT        | Required when applicable |
| audit_event_id  | BIGINT      | Optional foreign key     |

Status history is append-only.

## 7. Project-Code Generation

### 7.1 Format

The project identification code is:

text

Sequential Number-Sales Associate Initials-YYYY-MM-DD

Example:

text

101-JD-2026-09-19

The date is the Master Sales Record creation date.

### 7.2 Number sequence

The sequence:

Begins at 101.

Is generated by the database or a transaction-safe application service.

Is reserved during project or initial Master Sales Record creation.

Cannot be reused after cancellation.

Cannot be released back to the available pool after a failed or cancelled project creation.

Must remain unique permanently.

### 7.3 Sales Associate initials

The system must:

Require Sales Associate initials before code generation.

Validate the initials format.

Preserve the initials used in the original code.

Prevent revisions from changing the original project code.

Record the user who entered or confirmed the initials.

The exact initials format remains configurable but must be consistently enforced.

### 7.4 Project-code generation transaction

The following must occur transactionally:

Validate the Sales Associate.

Validate initials.

Reserve the next sequence number.

Determine the Master Sales Record creation date.

Construct the project code.

Check uniqueness.

Create the project.

Create the initial project audit event.

Commit the transaction.

If the transaction fails, the reserved sequence number must not be reused.

## 8. Duplicate Detection

### 8.1 Customer duplicate detection

The system should identify possible duplicate customers using combinations of:

Normalized customer name  
Phone number  
Email address  
Mailing address  
Existing customer number  
Duplicate-check key  
Possible matches must be shown before creation is completed.

## 8.2 Project-location duplicate detection

Before creating an active project, the system must check for:

Existing active project at the same normalized location  
Existing active project for the same customer and location  
Existing cancelled project at the location  
Existing completed project at the location  
Similar but not exact address matches

## 8.3 Duplicate warning behavior

If an active project exists at the same location:

Display the existing project code.  
Display only fields authorized for the current user.  
Require the user to review the possible duplicate.  
Require an explanation or correction.  
Prevent creation until the required explanation or correction is recorded.  
Require an authorized override where applicable.  
Log the warning, decision, and user.

## 8.4 Duplicate override

A duplicate-location override must include:

Existing project identifier  
New project identifier  
User requesting the override  
Date and time  
Explanation  
Approval or authorization result  
Approving user where required  
Related audit event  
A duplicate override must not delete, merge, or overwrite either project.

# 9. Customer and Project Search

## 9.1 Search fields

Authorized users may search by:

Customer name  
Phone number  
Email address  
Project identification code  
Street address  
City  
State  
ZIP code  
Sales Associate  
Technician

Project status

Date range

Customer number

Additional search fields will become available in later sprints:

Serial number

Brand

Model

Contract status

Invoice status

Certificate status

Payment status

Vendor

Payable

Document type

## 9.2 Search behavior

Search must:

Require authentication.

Apply record-level authorization.

Apply field-level authorization.

Return only records the user is permitted to view.

Prevent unauthorized data inference through counts or autocomplete.

Support exact and partial matching where appropriate.

Normalize phone, email, ZIP, and address search values.

Record the search as an audit event.

Record whether restricted results were requested.

Support pagination.

Display the user's permitted fields only.

## 9.3 Search result fields

The basic project search result should include:

Project identification code

Customer name

Project location

Assigned Sales Associate

Assigned Technician where authorized

Current project status

Creation date

Last activity date

Warning or exception indicator where authorized

Financial amounts, commissions, profitability, and payroll information must not appear unless separately authorized.

## 9.4 Search sorting

Default sorting:

Most recently active project

Most recently modified record

Project identification code

The system must provide authorized sorting by:

Customer name

Project code  
Location  
Status  
Sales Associate  
Creation date  
Last activity date  
10. Access Rules  
Sales Associate  
May:

Create customers  
Create projects  
View assigned customers  
View assigned projects  
Edit unapproved assigned project information  
Search within authorized assigned records  
May not:

View another Sales Associate's commission information  
View unauthorized projects  
View restricted financial amounts  
Override duplicate-location restrictions without authorization  
Edit approved project information  
Sales Manager and Comptroller  
May:

View all customers and projects  
Create and edit authorized records  
Resolve duplicate-location exceptions  
Assign or reassign authorized users  
Search all business records  
View all project and customer data  
Installation Manager  
May:

View installation-related projects  
View assigned or authorized project locations  
Search authorized projects  
Assign or review installation personnel where permitted  
Technician  
May:

View assigned projects  
View assigned project locations  
Search assigned projects  
View approved project information needed for technical work  
Accounts Payable Associate  
May:

View customer and project fields required for payable duties  
Search authorized project records  
View project codes and related payable references  
Database Administrator

May:

- Search authorized records
- Maintain approved database functions
- Perform approved data-quality operations
- The Database Administrator does not automatically receive unrestricted business-data access.

## 11. Customer and Project Forms

### 11.1 Customer form

The customer form must support:

- Create new customer
- Save as draft where applicable
- Add multiple phone numbers
- Add multiple email addresses
- Add multiple contacts
- Set preferred contact method
- Identify possible duplicates
- Save customer
- Edit unapproved customer information
- Retire customer through authorized action
- Clear form
- Cancel form

### 11.2 Project form

The project form must support:

- Select or create customer
- Enter project location
- Select Sales Associate
- Enter and validate Sales Associate initials
- Identify possible duplicate location
- Require duplicate explanation or correction
- Create project code
- Save project
- Save draft where applicable
- Assign technicians later through authorized workflow
- View project history
- Cancel form

### 11.3 Form validation

The system must validate:

- Required customer fields
- Required location fields
- Valid state code
- Valid ZIP-code format
- Valid phone format
- Valid email format
- Valid Sales Associate
- Valid initials
- Duplicate customer warning
- Duplicate active project warning
- Project-code uniqueness
- Authorized user scope

Record status  
Required explanation for duplicate override  
Required cancellation reason  
12. Audit Events  
Sprint 04 must create audit events for:

Customer events  
Customer created  
Customer viewed  
Customer edited  
Customer retired  
Possible duplicate identified  
Duplicate accepted  
Duplicate rejected  
Contact created  
Contact edited  
Contact retired  
Location events  
Location created  
Location viewed  
Location edited  
Location duplicate warning  
Location duplicate override  
Location retired  
Project events  
Project creation started  
Project-code number reserved  
Project created  
Project viewed  
Project edited  
Project assigned  
Project reassigned  
Project status changed  
Project cancelled  
Project duplicate warning  
Project duplicate override  
Project search  
Restricted project-access attempt  
Audit events must include:

User  
Role  
Session  
Resource  
Resource identifier  
Project or customer association  
Action  
Outcome  
Reason where applicable  
Previous and new status where applicable  
Audit-chain references  
13. Sprint 04 Decisions  
D-04-001 — Customer and project separation

Customers and projects are separate entities.

A customer may have multiple projects.

D-04-002 — Project location separation

Project location is a separate entity from the customer mailing address.

A project location includes distinct street, city, state, and ZIP fields.

D-04-003 — Project-code permanence

A project code is permanent once assigned and cannot be reused, including after cancellation.

D-04-004 — Project-code sequence

The project-code sequence begins at 101 and is reserved transactionally.

D-04-005 — Location duplicate behavior

An active project at the same customer location triggers a warning and requires an explanation or correction before another active project can be created.

D-04-006 — Duplicate preservation

Duplicate projects and customers are not silently merged or deleted.

Any later merge process must preserve both original records and create a separate audit trail.

D-04-007 — Search authorization

Search applies the same role, record, and field permissions as ordinary record viewing.

D-04-008 — Historical status preservation

Project-status history is append-only and is not replaced when the current status changes.

D-04-009 — State data

Project location state is stored as a controlled state code and is reserved for later automatic state-license-artwork selection.

D-04-010 — Sprint 05 dependency

The project may be created before the full Master Sales Record is implemented, but the project must not be treated as an approved sale until Sprint 05 and later lifecycle controls are complete.

#### 14. Sprint 04 Acceptance Criteria

Sprint 04 is accepted when:

A customer can be created and saved.

A customer can have multiple phone numbers.

A customer can have multiple email addresses.

A customer can have multiple contacts.

A project can be linked to exactly one customer.

A customer can have multiple projects.

A project location stores separate street, city, state, and ZIP fields.

A project can be linked to exactly one project location.

The location state is stored as a controlled value.

Possible duplicate customers are identified before creation.

Possible duplicate project locations are identified before active-project creation.

An active project at the same location triggers a warning.

A duplicate-location warning cannot be bypassed without an explanation or correction.  
Authorized duplicate overrides are recorded.  
Duplicate records are not silently merged or deleted.  
The project-code sequence begins at 101.  
Project codes use the approved format.  
Project codes are unique.  
Sequence numbers are reserved transactionally.  
Cancelled project codes cannot be reused.  
Project revisions preserve the original project code.  
A project has an assigned Sales Associate.  
A project may have multiple assigned Technicians.  
Project assignments are effective-dated.  
Project-status history is append-only.  
Authorized users can search customers and projects.  
Unauthorized users cannot use search to discover restricted records.  
Search results respect role and field permissions.  
Search activity is logged.  
Customer and project views are logged.  
Project-code reservation is logged.  
Duplicate warnings and overrides are logged.  
Required fields are validated before save.  
Invalid state, ZIP, email, phone, and project-code data are rejected.  
Retired customer, location, and project records remain historically retrievable according to permissions.  
No project is treated as approved solely because it exists in Sprint 04.

#### 15. Sprint 04 Test Cases

##### Customer tests

Create a valid customer.  
Reject a customer with no required name.  
Add multiple phone numbers.  
Add multiple email addresses.  
Add multiple contacts.  
Set a primary contact.  
Prevent conflicting active primary contacts.  
Identify a possible duplicate by normalized name and phone.  
Retire a customer with a required reason.  
Preserve the customer's history after retirement.

##### Location tests

Create a valid location.  
Reject a location without street, city, state, or ZIP.  
Reject an invalid state code.  
Normalize equivalent address values.  
Match equivalent normalized locations.  
Preserve the original entered address.  
Identify a possible active-project location duplicate.

##### Project-code tests

Create the first project with sequence number 101.  
Create a second project with sequence number 102.  
Validate the initials portion.  
Validate the date portion.  
Reject a duplicate project code.  
Cancel a project and confirm its code cannot be reused.  
Create a later revision and confirm the code remains unchanged.

Simulate concurrent project creation and confirm unique sequence allocation.  
Simulate failed project creation and confirm the reserved number is not reused.

#### Access tests

Sales Associate sees assigned projects.

Sales Associate cannot see unassigned projects.

Technician sees assigned projects only.

Accounts Payable Associate sees only authorized project fields.

Installation Manager sees authorized installation-related projects.

Sales Manager sees all projects.

Comptroller sees all projects.

Database Administrator receives only authorized access.

Unauthorized search results are excluded.

Unauthorized fields are not returned through search, print, or export.

#### Audit tests

Customer creation creates an audit event.

Project creation creates an audit event.

Project-code reservation creates an audit event.

Project assignment creates an audit event.

Duplicate warning creates an audit event.

Duplicate override creates an audit event.

Restricted-access attempt creates an audit event.

Search creates an audit event.

Project cancellation creates an audit event.

#### 16. Sprint 04 Open Questions

These implementation questions do not change approved requirements:

What exact customer-number format will be used?

What address-normalization library or service will be used?

What level of fuzzy address matching is required?

Which duplicate matches are warnings versus hard blocks?

Who may approve a duplicate-location override?

What exact Sales Associate initials format will be enforced?

Should project creation reserve a number before or after customer confirmation?

What additional project statuses are needed before Sprint 06?

What customer-merge process, if any, will be implemented later?

What states and territories must be included in the initial state-code reference table?

What search result limits and pagination sizes are appropriate?

What fields may be printed by each role?

#### 17. Sprint 05 Handoff

Sprint 05 will implement the Master Sales Record.

Sprint 05 must use the Sprint 04 structures for:

Customer

Customer contacts

Project

Project location

Assigned Sales Associate

Project code

Project status

Project audit history

Sprint 05 must preserve:

The original project identification code  
The project-to-customer relationship  
The project location  
The assigned Sales Associate  
Existing project assignments  
Existing duplicate warnings and overrides  
All customer and project audit events  
Sprint 05 must not create a second project identifier for the same project.

## 18. Historical Record

Supersedes  
None.

Preserves

GECC-00-project-baseline.md  
GECC-01-data-model.md  
GECC-02-auth-and-permissions.md  
GECC-03-audit-log.md  
Sprint 00 decisions  
Sprint 01 decisions  
Sprint 02 decisions  
Sprint 03 decisions  
Approved access restrictions  
Approved project-code requirements  
Approved duplicate-location requirements  
Changes introduced by this bundle  
Defines the customer and contact data structures.  
Defines project-location storage and normalization.  
Defines project and assignment structures.  
Defines permanent project-code generation behavior.  
Defines customer and project duplicate detection.  
Defines authorized customer and project search.  
Defines Sprint 04 audit events.  
Defines Sprint 05 Master Sales Record handoff.  
===== END FILE: GECC-04-Customers-And-Projects.md =====  
=== BEGIN FILE: GECC-05-Master-Sales-Record.md ===  
# GECC-05-master-sales-record.md

# GECC Sales Back Office System — Sprint 05 Master Sales Record

## Bundle Metadata

- Bundle name: GECC-05-master-sales-record.md
- Project: GECC Sales Back Office System
- Company: Go Ecco Climate Control
- Company abbreviation: GECC
- Bundle type: Sprint 05 design and implementation bundle
- Status: Approved design; ready for implementation
- Created date: 2026-09-20
- Previous bundle: GECC-04-customers-and-projects.md
- Next planned bundle: GECC-06-workflow-state-machine.md
- Database: PostgreSQL
- Primary-key strategy: Numeric BIGINT identity keys

- Business timezone: America/New\_York
- System of record: PostgreSQL relational database
- Audit model: Append-only, hash-chained, permanently retained

---

## ## 1. Sprint Objective

Implement the Master Sales Record as the authoritative project record for:

- Customer information
- Project information
- Project location
- Sales Associate assignment
- Scope of work
- Equipment
- Serial numbers
- Retail price
- Payment option
- Project costs
- Contract data
- Invoice data
- Certificate data
- Approval status
- Document-version references
- Revision reason
- Revision history

The Master Sales Record must allow Sales Associates to enter project information once and provide the controlled source data for future invoices, contracts, certificates, commissions, payments, and financial reports.

---

## ## 2. Sprint 05 Deliverables

Sprint 05 must produce:

1. Master Sales Record form
2. Draft Master Sales Record behavior
3. Required-field validation
4. Customer and project consistency validation
5. Scope-of-work selections
6. Equipment records
7. Serial-number records
8. Pricing fields
9. Cost fields
10. Payment-option fields
11. Versioned Master Sales Records
12. Approval-ready records
13. Approval history
14. Revision reasons
15. Current and superseded version handling

16. Controlled access to pricing and costs
17. Master Sales Record search and display
18. Audit integration
19. Database acceptance tests
20. Sprint 06 workflow handoff

---

## ## 3. Requirements Carried Forward

The implementation must preserve these approved requirements:

- Each project has one Master Sales Record.
- The Master Sales Record is the authoritative source for project information.
- The Master Sales Record is linked to the customer and project.
- A project may include multiple pieces of equipment.
- A project may include multiple serial numbers.
- The Sales Associate creates the initial record.
- Unapproved information may be updated by authorized users.
- A Sales Manager or Comptroller reviews and approves the record.
- The authorized approver enters or confirms the retail price.
- One authorized approval is sufficient.
- Approved records are locked.
- Authorized reopening creates a new version.
- Original versions remain permanently preserved.
- Scope changes create a new Master Sales Record version.
- Revised documents use the revised version.
- Original documents and signatures remain preserved.
- The original project code remains unchanged through revisions.
- Documents must use the approved Master Sales Record.
- Unauthorized users may not edit approved retail price or approved costs.
- All material changes are auditable.

---

## ## 4. Master Sales Record Structure

### ### 4.1 `master\_sales\_record`

This table identifies the logical Master Sales Record for a project.

| Field                | Type        | Requirement                                    |
|----------------------|-------------|------------------------------------------------|
| ---                  | ---         | ---                                            |
| `id`                 | BIGINT      | Primary key                                    |
| `project_id`         | BIGINT      | Required foreign key, unique                   |
| `current_version_id` | BIGINT      | Required after initial version creation        |
| `record_status`      | VARCHAR(30) | Draft, active, approved, superseded, cancelled |
| `created_at`         | TIMESTAMPTZ | Required                                       |
| `created_by`         | BIGINT      | Required foreign key                           |
| `updated_at`         | TIMESTAMPTZ | Required                                       |
| `updated_by`         | BIGINT      | Required foreign key                           |
| `is_active`          | BOOLEAN     | Required                                       |
| `retired_at`         | TIMESTAMPTZ | Optional                                       |

| `retired\_by` | BIGINT | Optional foreign key |  
| `retirement\_reason` | TEXT | Optional |

Constraint:

```text

UNIQUE(project_id)

A project may have only one logical Master Sales Record.

4.2 master_sales_record_version

Field Type Requirement

id BIGINT Primary key

master_sales_record_id BIGINT Required foreign key

project_id BIGINT Required foreign key

customer_id BIGINT Required foreign key

version_number INTEGER Required

version_status VARCHAR(30) Draft, submitted, approved, rejected, superseded, cancelled

created_at TIMESTAMPTZ Required

created_by BIGINT Required foreign key

revision_reason TEXT Required for version greater than 1

supersedes_version_id BIGINT Optional foreign key

assigned_sales_associate_id BIGINT Required foreign key

retail_price NUMERIC(19,4) Required before approval

payment_option VARCHAR(40) Required

scope_summary TEXT Required

approval_status VARCHAR(30) Required

approved_by BIGINT Optional foreign key

approved_at TIMESTAMPTZ Optional

rejected_by BIGINT Optional foreign key

rejected_at TIMESTAMPTZ Optional

rejection_reason TEXT Optional

source_version_id BIGINT Optional foreign key

document_data_version INTEGER Required

record_hash VARCHAR(128) Optional integrity reference

Constraint:

text

UNIQUE(master_sales_record_id, version_number)

Only one version may be current for a logical Master Sales Record.

5. Master Sales Record Sections

The form must be divided into controlled sections.

5.1 Customer section

The form displays linked customer information:

Customer ID

Customer number

Customer name

Primary phone

Primary email

Authorized contacts

Customer information is retrieved from the Customer record.

The Master Sales Record must not create an independent conflicting copy of customer identity data.

5.2 Project section

The form displays linked project information:

Project ID

Project identification code

Project location

Assigned Sales Associate

Assigned Technician or Technicians

Current project status

Project creation date

The project identification code is read-only after creation.

5.3 Scope section

The form supports multiple scope selections:

Service

Repair

Install new

Remove old

Under Install New and Remove Old, the form supports multiple items:

Condenser coil

Evaporator coil

Refrigerant

Air handler

Furnace

Thermostat

Air ducts

Vents

A project may include more than one scope selection.

5.4 Equipment section

The form supports multiple equipment records.

Required fields may vary according to scope and equipment status:

Brand

Model

Function

Serial number

Equipment category

New, existing, removed, or installed status

Inventory reference

Notes

The form must support:

- Add equipment
- Edit unapproved equipment
- Remove an unapproved equipment row
- Record existing equipment
- Record equipment to be removed
- Record equipment to be installed
- Update serial numbers where authorized

5.5 Pricing section

The pricing section includes:

- Retail price
- Payment option
- Adjustment placeholder
- Pricing notes where authorized

Retail price must be entered or confirmed by:

Sales Manager, or
Comptroller

The Sales Associate may enter proposed pricing where permitted but may not approve the final retail price.

5.6 Cost section

The cost section includes:

- Equipment cost
- Material cost
- Dealer fee
- Lead cost
- Technician cost
- Damon cost
- Jerry cost

Initial default values:

text

Lead: \ \$3,000

Damon: \ \$500

Jerry: \ \$250

The default values apply to the project unless an authorized user creates a project-specific change.

A change to one project must not silently alter:

- Past projects
- Future projects
- Other projects
- Global defaults

5.7 Payment-option section

Initial supported payment methods:

- Bank cashier's check
- Bank wire transfer

The payment option identifies the planned method and does not constitute payment receipt or verification.

5.8 Document-data section

The Master Sales Record stores or references the controlled data needed for:

Invoice

Contract

Certificate

Document generation is implemented in Sprint 07. Sprint 05 must provide stable version references for later document generation.

6. Scope and Equipment Tables

6.1 msr_scope_selection

| Field | Type | Requirement |
|-------|------|-------------|
|-------|------|-------------|

| | | |
|----|--------|-------------|
| id | BIGINT | Primary key |
|----|--------|-------------|

| | | |
|----------------|--------|----------------------|
| msr_version_id | BIGINT | Required foreign key |
|----------------|--------|----------------------|

| | | |
|------------|-------------|----------|
| scope_code | VARCHAR(40) | Required |
|------------|-------------|----------|

| | | |
|-------------|-------------|----------|
| scope_group | VARCHAR(40) | Required |
|-------------|-------------|----------|

| | | |
|-------------|---------|----------|
| is_selected | BOOLEAN | Required |
|-------------|---------|----------|

| | | |
|-------|------|----------|
| notes | TEXT | Optional |
|-------|------|----------|

| | | |
|------------------------|---|----------|
| Standard audit columns | — | Required |
|------------------------|---|----------|

6.2 msr_equipment

| Field | Type | Requirement |
|-------|------|-------------|
|-------|------|-------------|

| | | |
|----|--------|-------------|
| id | BIGINT | Primary key |
|----|--------|-------------|

| | | |
|----------------|--------|----------------------|
| msr_version_id | BIGINT | Required foreign key |
|----------------|--------|----------------------|

| | | |
|------------|--------|----------------------|
| project_id | BIGINT | Required foreign key |
|------------|--------|----------------------|

| | | |
|--------------------|---------|----------|
| equipment_sequence | INTEGER | Required |
|--------------------|---------|----------|

| | | |
|-------|--------------|---------------------------|
| brand | VARCHAR(150) | Required where applicable |
|-------|--------------|---------------------------|

| | | |
|-------|--------------|---------------------------|
| model | VARCHAR(150) | Required where applicable |
|-------|--------------|---------------------------|

| | | |
|----------|--------------|----------|
| function | VARCHAR(100) | Required |
|----------|--------------|----------|

| | | |
|---------------|--------------|-------------------------------------|
| serial_number | VARCHAR(150) | Optional until required by workflow |
|---------------|--------------|-------------------------------------|

| | | |
|--------------------|-------------|----------|
| equipment_category | VARCHAR(80) | Required |
|--------------------|-------------|----------|

| | | |
|------------------|-------------|-----------------------------------|
| equipment_status | VARCHAR(40) | New, existing, removed, installed |
|------------------|-------------|-----------------------------------|

| | | |
|--------------|--------|----------------------|
| inventory_id | BIGINT | Optional foreign key |
|--------------|--------|----------------------|

| | | |
|-------|------|----------|
| notes | TEXT | Optional |
|-------|------|----------|

| | | |
|------------------------|---|----------|
| Standard audit columns | — | Required |
|------------------------|---|----------|

Constraint:

text

UNIQUE(msr_version_id, equipment_sequence)

6.3 Serial-number rules

Serial numbers must remain individually identifiable.

Duplicate serial numbers must be flagged.

A serial number may not be silently reassigned to a different project.

Serial-number changes after approval require authorized revision.

The original serial-number value remains available in history.

Missing serial numbers may be permitted in draft status only where the applicable scope does not yet require them.

7. Cost Tables

7.1 msr_cost_component

| Field | Type | Requirement |
|---------------------|---------------|----------------------|
| id | BIGINT | Primary key |
| msr_version_id | BIGINT | Required foreign key |
| cost_type | VARCHAR(40) | Required |
| amount | NUMERIC(19,4) | Required |
| default_amount | NUMERIC(19,4) | Optional |
| is_default_applied | BOOLEAN | Required |
| is_project_override | BOOLEAN | Required |
| entered_by | BIGINT | Required foreign key |
| entered_at | TIMESTAMPTZ | Required |
| approval_status | VARCHAR(30) | Required |
| approved_by | BIGINT | Optional foreign key |
| approved_at | TIMESTAMPTZ | Optional |
| notes | TEXT | Optional |

7.2 Cost permissions

Equipment Cost, Material Cost, and Dealer Fee may be entered by:

Sales Manager
 Installation Manager
 Comptroller
 Technician Cost may be entered by:

Sales Manager
 Installation Manager
 Comptroller
 Lead Cost, Damon Cost, and Jerry Cost may be edited for an individual project by authorized users.

Sprint 05 must enforce field permissions but Sprint 06 will enforce the complete lifecycle timing.

8. Draft Behavior

8.1 Draft creation

A Sales Associate may create a draft Master Sales Record for an assigned project.

Draft creation must require:

Customer
 Project
 Project location
 Assigned Sales Associate
 Project code
 Initial scope or stated scope-pending reason

8.2 Draft editing

Before approval, authorized users may edit permitted fields.

Draft edits must:

Update the draft version.
 Preserve changed-field history.
 Update updated_at and updated_by.
 Generate an audit event.
 Revalidate the complete record before submission.

8.3 Save as draft

The form must support:

- Save draft
- Resume draft
- Cancel entry
- Clear unsaved form
- View validation errors
- Submit for review

A draft is not an approved sale and cannot generate final controlled documents.

8.4 Draft deletion

Draft records must not be hard-deleted without an authorized retirement or cancellation action.

The system must preserve:

- Creator
- Creation time
- Cancellation or retirement reason
- Audit history

9. Submission and Approval

9.1 Submission for review

A Sales Associate may submit a complete draft for review.

Submission requires validation of:

- Required customer information
- Project consistency
- Location consistency
- Project-code consistency
- Assigned Sales Associate
- Scope
- Equipment requirements
- Pricing
- Costs
- Payment option
- Duplicate data
- Required explanations
- Record version

9.2 Approval authority

The approved core lifecycle requires approval by:

- Sales Manager, or
- Comptroller

One authorized approval is sufficient.

The implementation must retain a flexible approval structure so the later workflow sprint can represent any approved applicable workflow distinction without weakening the core requirement.

9.3 Approval result

An approver may:

- Approve
- Reject

Return for correction
Approval must record:

Exact Master Sales Record version
Approver
Approver role
Date and time
Approval result
Approval notes
Related audit event
9.4 Approved record behavior
After approval:

The approved version becomes immutable.
Ordinary edits are rejected.
The current version reference is updated.
Later changes require authorized reopening or a new revision.
Documents may be generated from the approved version in later sprints.
The record cannot be silently overwritten.

10. Versioning and Revision

10.1 New version creation
A new version is required when:

An approved record is reopened.
A signed or approved project changes scope.
Approved pricing changes.
Approved costs change.
Contract data changes.
Invoice data changes.
Certificate data changes.
An authorized correction is required.

10.2 Revision requirements
Every revision must include:

New version number
Prior version reference
Revision reason
Revising user
Revision date and time
Changed fields
New status
Approval state
Related audit event
10.3 Version relationships
The system must preserve:

text

Version 1
└─ Version 2
 └─ Version 3

Each version must remain independently retrievable.

10.4 Current version

Only one version may be marked current.

The current version must be:

Linked from master_sales_record.current_version_id

Linked from project.current_msr_version_id

Clearly marked in user interfaces

Used by later document generation only when its status permits generation

10.5 Superseded versions

A superseded version:

Remains read-only

Remains searchable according to permissions

Remains linked to its documents and approvals

Cannot be edited

Cannot be used as the current source for new documents

Remains available for audit and historical reporting

11. Validation Rules

Before saving a draft:

Required draft fields must be present.

Foreign-key references must be valid.

The user must have permission to create or edit the record.

The project must exist.

The customer must match the project.

The project location must match the project.

The assigned Sales Associate must be valid.

Scope values must come from controlled values.

Equipment values must be valid.

Monetary values must be non-negative unless an approved accounting rule permits otherwise.

Payment option must be supported.

Duplicate serial numbers must be flagged.

Before submission for approval:

Required fields must be complete.

Customer consistency must pass.

Project consistency must pass.

Project-code consistency must pass.

Scope consistency must pass.

Equipment and serial-number requirements must pass.

Retail price must be present.

Costs must be present or explicitly marked not applicable.

Payment option must be present.

Required duplicate explanations must be present.

No prohibited draft errors may remain.

Before approval:

The approver must be authorized.

The record must still be the current submitted version.

The record must not already be approved.

The record must not be superseded.

The record must not be cancelled.

The approval must reference the exact version being approved.

12. Access Rules

Sales Associate

May:

Create Master Sales Records for assigned projects.

Edit unapproved records within assigned projects.

View generated draft information.

Submit records for review.

Review the generated contract in later sprints.

View commission information related only to their own sales.

May not:

Approve a Master Sales Record.

Edit an approved Master Sales Record.

Change approved retail price.

Change approved costs.

Approve commission or payroll reports.

View another Sales Associate's commissions.

Sales Manager

May:

View all Master Sales Records.

Create or edit authorized records.

Enter or confirm retail price.

Enter or revise permitted costs.

Approve Master Sales Records.

Reopen approved records.

Approve applicable corrections.

Comptroller

May:

View all Master Sales Records.

Create or edit authorized records.

Enter or confirm retail price.

Enter or revise permitted costs.

Approve Master Sales Records.

Reopen approved records.

Approve applicable corrections.

Installation Manager

May:

View installation-related Master Sales Records.

Enter or approve installation-related cost information according to the applicable authorization.

View approved scope and equipment information.

The core sales-approval lifecycle remains Sales Manager or Comptroller approval unless a later approved decision explicitly changes it.

Technician

May:

View approved scope of work for assigned projects.

View required equipment information.
Update authorized serial-number and completion information in later workflow stages.
Propose scope changes.
May not:

Change retail price.
Change approved costs.
Change contract terms.
Approve the Master Sales Record.
Accounts Payable Associate
May:

View Master Sales Record fields necessary for payable duties.
View project, vendor, cost-category, and payable references where authorized.
Database Administrator
May:

Maintain approved database functions.
Search authorized records.
Perform approved data-quality maintenance.
The Database Administrator does not automatically receive Master Sales Record approval authority.

13. Audit Events

Sprint 05 must create audit events for:

Master Sales Record draft created
Draft viewed
Draft edited
Draft saved
Draft resumed
Draft cancelled
Draft submitted for review
Validation failure
Master Sales Record approved
Master Sales Record rejected
Master Sales Record returned for correction
Master Sales Record reopened
New version created
Version superseded
Revision reason entered
Scope changed
Equipment added
Equipment changed
Serial number added
Serial number changed
Retail price entered
Retail price changed
Cost entered
Cost changed
Payment option changed
Unauthorized edit attempt
Unauthorized approval attempt

Restricted Master Sales Record access attempt

Each event must include:

User

Role

Session

Project

Customer

Master Sales Record

Version

Action

Outcome

Reason where applicable

Previous and new status where applicable

Audit-chain references

14. Sprint 05 Decisions

D-05-001 — Master Sales Record authority

The Master Sales Record is the authoritative source of project information for downstream documents, payments, commissions, and reports.

D-05-002 — One logical record per project

Each project has one logical Master Sales Record with one or more preserved versions.

D-05-003 — Immutable approved versions

Approved Master Sales Record versions are immutable.

D-05-004 — Revision behavior

Changes to approved records create new versions and preserve previous versions.

D-05-005 — Core approval authority

The core approval lifecycle requires approval by either the Sales Manager or Comptroller.

One authorized approval is sufficient.

D-05-006 — Approval version binding

Approvals must reference the exact Master Sales Record version being approved.

D-05-007 — Project-code preservation

Master Sales Record revisions do not create a new project code.

D-05-008 — Default-cost isolation

Changes to Lead, Damon, Jerry, or other project costs apply only to the indicated project version unless a separately approved configuration change is made.

D-05-009 — Document-generation dependency

Sprint 05 stores stable version references for documents, but document generation itself is implemented in Sprint 07.

D-05-010 — Calculation separation

Sprint 05 stores source pricing and cost inputs. Final profitability snapshots and accounting calculations remain subject to later financial-reporting design.

15. Sprint 05 Acceptance Criteria

Sprint 05 is accepted when:

A Sales Associate can create a Master Sales Record for an assigned project.

A Master Sales Record is linked to exactly one project.

A project has only one logical Master Sales Record.

The customer and project relationships are validated.

The project code is inherited from the project and cannot be changed through the Master Sales Record.

Scope selections support service, repair, install new, and remove old.

Install-new and remove-old sub-items are supported.

Multiple equipment records can be added.

Multiple serial numbers can be stored.

Duplicate serial numbers are detected.

Retail price is stored as a fixed-precision amount.

Payment option supports cashier's check and bank wire.

Required cost categories are supported.

Initial fixed default costs are represented.

Project-specific cost overrides do not change other projects.

Draft records can be saved and resumed.

Draft records can be submitted for review.

Required fields are validated before submission.

Inconsistent customer, project, location, and project-code values are rejected.

An authorized Sales Manager or Comptroller can approve a submitted version.

One authorized approval is sufficient.

An unauthorized user cannot approve the record.

An approved version cannot be silently edited.

An approved version can be superseded only through an authorized revision process.

Revision reason, user, date, and time are preserved.

Previous versions remain retrievable.

Only one current version exists.

Superseded versions cannot be used as the current source for new documents.

Access to pricing and cost fields is role-controlled.

Sales Associates cannot view another Sales Associate's commission data.

All material Master Sales Record actions create audit events.

Failed and unauthorized actions create audit events.

The data model is ready for Sprint 06 lifecycle and state-machine enforcement.

16. Sprint 05 Test Cases

Creation and draft tests

Create a valid Master Sales Record.

Reject creation for an unauthorized project.

Save an incomplete draft where permitted.

Resume a saved draft.

Reject invalid customer-project combinations.

Reject invalid project-location combinations.

Reject unsupported payment options.

Reject invalid monetary values.

Clear a draft form without saving unsaved changes.

Scope and equipment tests

Select service.

Select repair.

Select install new.

Select remove old.

Select multiple install-new sub-items.

Select multiple remove-old sub-items.
Add multiple equipment records.
Add multiple serial numbers.
Reject duplicate serial numbers.
Preserve equipment history after revision.
Pricing and cost tests
Enter proposed retail price.
Confirm retail price as Sales Manager.
Confirm retail price as Comptroller.
Reject Sales Associate approval.
Enter permitted equipment cost.
Enter permitted material cost.
Enter permitted dealer fee.
Apply initial fixed defaults.
Override a fixed default for one project.
Confirm the override does not change another project.
Approval tests
Submit a valid record for review.
Reject submission with missing required data.
Approve as Sales Manager.
Approve as Comptroller.
Reject approval by Sales Associate.
Reject approval by Technician.
Reject approval of a superseded version.
Reject duplicate approval of the same version.
Preserve approver and timestamp.
Versioning tests
Create Version 1.
Approve Version 1.
Reopen through an authorized action.
Create Version 2.
Preserve Version 1 unchanged.
Record revision reason.
Mark Version 1 superseded.
Mark Version 2 current.
Confirm the project code remains unchanged.
Confirm Version 1 remains searchable according to permission.
Security and audit tests
Unauthorized user cannot view restricted cost fields.
Unauthorized user cannot edit approved fields.
Unauthorized user cannot approve a record.
Unauthorized user cannot reopen a record.
Draft creation is audited.
Approval is audited.
Revision is audited.
Serial-number changes are audited.
Unauthorized attempts are audited.
Failed audit creation prevents the required business action from succeeding.

17. Sprint 05 Open Questions

These implementation questions do not change approved requirements:

What exact fields are mandatory for each scope and equipment category?
Which equipment categories require serial numbers before approval?

What exact initials format will be used for Sales Associates?
What cost fields may the Installation Manager edit before installation?
What fields may be entered by the Comptroller when acting as a substitute signer later?
What exact return-for-correction workflow will Sprint 06 implement?
What approval reason values should be standardized?
What calculation-rule version should be stored with pricing and cost inputs?
What project-specific cost override process requires Sales Manager approval?
What Master Sales Record fields must be copied into document-generation staging tables?

18. Sprint 06 Handoff

Sprint 06 will implement the controlled lifecycle and task workflow.

Sprint 06 must use the following Sprint 05 structures:

Master Sales Record
Master Sales Record version
Scope selections
Equipment
Serial numbers
Pricing
Cost components
Payment option
Approval status
Revision reason
Current-version reference
Approval history
Sprint 06 must enforce:

Required state transitions
Approval sequencing
Installation task creation
Three-day installation restriction
Scope-change process
Authorized reopening
New-version creation
Revised approval and signature requirements
Sprint 06 must not overwrite Sprint 05 versions or bypass Master Sales Record immutability.

19. Historical Record

Supersedes
None.

Preserves
GECC-00-project-baseline.md
GECC-01-data-model.md
GECC-02-auth-and-permissions.md
GECC-03-audit-log.md
GECC-04-customers-and-projects.md
Sprint 00 decisions
Sprint 01 decisions
Sprint 02 decisions
Sprint 03 decisions
Sprint 04 decisions
Approved project-code requirements

Approved lifecycle requirements
Approved versioning and audit requirements
Changes introduced by this bundle
Defines the Master Sales Record structure.
Defines draft, submission, approval, and revision behavior.
Defines scope and equipment data.
Defines pricing, payment-option, and cost inputs.
Defines Master Sales Record versioning.
Defines access restrictions for pricing and cost data.
Defines Sprint 05 audit events.
Defines Sprint 06 workflow handoff.
===== END FILE: GECC-05-Master-Sales-Record.md =====
=== BEGIN FILE: GECC-06-workflow-state-machine.md ===
GECC-06-workflow-state-machine.md

GECC Sales Back Office System — Sprint 06 Controlled Lifecycle and Task Workflow

Bundle Metadata

- Bundle name: GECC-06-workflow-state-machine.md
- Project: GECC Sales Back Office System
- Company: Go Ecco Climate Control
- Company abbreviation: GECC
- Bundle type: Sprint 06 design and implementation bundle
- Status: Approved design; ready for implementation
- Created date: 2026-09-20
- Previous bundle: GECC-05-master-sales-record.md
- Next planned bundle: GECC-07-document-generation.md
- Database: PostgreSQL
- Primary-key strategy: Numeric BIGINT identity keys
- Business timezone: America/New_York
- System of record: PostgreSQL relational database
- Audit model: Append-only, hash-chained, permanently retained

1. Sprint Objective

Implement the controlled project lifecycle and task workflow required to prevent users from:

- Skipping required steps
- Performing actions out of order
- Installing before the required waiting period
- Changing approved records without authorization
- Continuing changed-scope work before approval and signatures
- Marking projects complete before all required conditions are satisfied

The system must create, assign, track, complete, and audit workflow tasks.

2. Sprint 06 Deliverables

Sprint 06 must produce:

1. Project state machine
2. Master Sales Record state transitions
3. Document and signature dependency checks
4. Installation task workflow
5. Three-day installation restriction
6. Waiting-period calculation
7. Scope-change workflow
8. Authorized reopening workflow
9. Version-creation workflow
10. Approval and correction tasks
11. Task assignment and due dates
12. Exception and override handling
13. State-transition audit events
14. Workflow dashboard data
15. Workflow acceptance tests
16. Sprint 07 document-generation handoff

3. Lifecycle Principles

The workflow must preserve these approved requirements:

- Users cannot skip required steps.
- Approved records are locked.
- Authorized reopening creates a new version.
- Original versions remain preserved.
- Documents use the approved Master Sales Record.
- The customer signs the contract before GECC signs it.
- The GECC signer signs the invoice before the customer signs it.
- Installation cannot occur earlier than three days after the customer signs both the contract and invoice.
- The certificate is signed by the customer before the Installation Manager signs it.
- The Comptroller may sign the certificate only when the Installation Manager is unavailable and must record the reason.
- Payment verification is required before project completion.
- Scope changes require review, approval, revised pricing and costs where applicable, a new Master Sales Record version, revised documents, and a new signature sequence.
- Original documents and records remain permanently preserved.

4. Project State Machine

4.1 State list

The initial project lifecycle uses these states:

```text

draft

sales\\_record\\_pending\\_review

sales\\_record\\_rejected  
 sales\\_record\\_approved  
 contract\\_pending\\_customer\\_signature  
 contract\\_pending\\_gecc\\_signature  
 contract\\_signed  
 invoice\\_pending\\_gecc\\_signature  
 invoice\\_pending\\_customer\\_signature  
 invoice\\_signed  
 installation\\_waiting\\_period  
 installation\\_scheduling\\_required  
 installation\\_scheduled  
 installation\\_in\\_progress  
 installation\\_completed  
 certificate\\_pending\\_customer\\_signature  
 certificate\\_pending\\_gecc\\_signature  
 certificate\\_signed  
 payment\\_pending  
 payment\\_verification\\_pending  
 adjustment\\_pending\\_approval  
 project\\_ready\\_for\\_completion  
 completed  
 scope\\_change\\_pending  
 scope\\_change\\_approved  
 scope\\_change\\_rejected  
 revision\\_pending\\_approval  
 cancelled  
 exception\\_review

Some document-specific states may be stored on document records while the project maintains the overall business state.

## 4.2 State descriptions

### State Meaning

draft Project or Master Sales Record is being prepared  
 sales\_record\_pending\_review Record is submitted for authorized review  
 sales\_record\_rejected Record was rejected or returned for correction  
 sales\_record\_approved Master Sales Record is approved and locked  
 contract\_pending\_customer\_signature Contract is awaiting customer signature  
 contract\_pending\_gecc\_signature Contract is awaiting GECC signature  
 contract\_signed Contract has all required signatures  
 invoice\_pending\_gecc\_signature Invoice is awaiting GECC signature  
 invoice\_pending\_customer\_signature Invoice is awaiting customer signature  
 invoice\_signed Invoice has all required signatures  
 installation\_waiting\_period Required three-day waiting period is active  
 installation\_scheduling\_required Installation Manager scheduling task is open  
 installation\_scheduled Installation date has been assigned  
 installation\_in\_progress Work has started  
 installation\_completed Technician has recorded completion  
 certificate\_pending\_customer\_signature Certificate is awaiting customer signature  
 certificate\_pending\_gecc\_signature Certificate is awaiting GECC signature  
 certificate\_signed Certificate has all required signatures  
 payment\_pending Payment has not yet been recorded  
 payment\_verification\_pending Payment is recorded but not verified  
 adjustment\_pending\_approval Credit, refund, or adjustment requires approval

project\_ready\_for\_completion All completion prerequisites are satisfied  
completed Project is complete and eligible for commissions  
scope\_change\_pending Technician has proposed a scope change  
scope\_change\_approved Scope change has been approved and revision work may proceed  
scope\_change\_rejected Scope change was rejected  
revision\_pending\_approval Revised Master Sales Record awaits approval  
cancelled Project is cancelled and cannot reuse its project code  
exception\_review Authorized exception review is active

## 5. Allowed State Transitions

### 5.1 Initial sales process

text

draft

→ sales\\_record\\_pending\\_review

sales\\_record\\_pending\\_review

→ sales\\_record\\_approved

→ sales\\_record\\_rejected

sales\\_record\\_rejected

→ draft

→ cancelled

sales\\_record\\_approved

→ contract\\_pending\\_customer\\_signature

### 5.2 Contract process

text

contract\\_pending\\_customer\\_signature

→ contract\\_pending\\_gecc\\_signature

→ contract\\_pending\\_customer\\_signature

→ cancelled

contract\\_pending\\_gecc\\_signature

→ contract\\_signed

→ contract\\_pending\\_customer\\_signature

→ cancelled

The customer must sign the contract before the GECC signer signs it.

### 5.3 Invoice process

text

contract\\_signed

→ invoice\\_pending\\_gecc\\_signature

invoice\\_pending\\_gecc\\_signature

→ invoice\\_pending\\_customer\\_signature

→ cancelled

invoice\\_pending\\_customer\\_signature

- invoice\\_signed
- invoice\\_pending\\_gecc\\_signature
- cancelled

The GECC signer must sign the invoice before the customer signs it.

#### 5.4 Installation process

text

invoice\\_signed

- installation\\_waiting\\_period

installation\\_waiting\\_period

- installation\\_scheduling\\_required
- exception\\_review

installation\\_scheduling\\_required

- installation\\_scheduled

installation\\_scheduled

- installation\\_in\\_progress
- installation\\_scheduling\\_required

installation\\_in\\_progress

- installation\\_completed
- scope\\_change\\_pending

#### 5.5 Certificate process

text

installation\\_completed

- certificate\\_pending\\_customer\\_signature

certificate\\_pending\\_customer\\_signature

- certificate\\_pending\\_gecc\\_signature

certificate\\_pending\\_gecc\\_signature

- certificate\\_signed

The customer must sign the certificate before the authorized GECC signer signs it.

#### 5.6 Payment and completion process

text

certificate\\_signed

- payment\\_pending
- payment\\_verification\\_pending

payment\\_pending

- payment\\_verification\\_pending

payment\\_verification\\_pending

- adjustment\\_pending\\_approval

→ project\\_ready\\_for\\_completion

adjustment\\_pending\\_approval

→ payment\\_verification\\_pending

→ project\\_ready\\_for\\_completion

A project may enter project\_ready\_for\_completion only when all completion conditions are satisfied.

text

project\\_ready\\_for\\_completion

→ completed

## 5.7 Scope-change process

text

installation\\_in\\_progress

→ scope\\_change\\_pending

scope\\_change\\_pending

→ scope\\_change\\_approved

→ scope\\_change\\_rejected

scope\\_change\\_rejected

→ installation\\_in\\_progress

scope\\_change\\_approved

→ revision\\_pending\\_approval

revision\\_pending\\_approval

→ sales\\_record\\_approved

→ sales\\_record\\_rejected

After the revised Master Sales Record is approved, revised documents must enter a new required signature sequence.

## 5.8 Reopening process

An approved or signed project may be reopened only by an authorized user.

text

approved\\_or\\_signed\\_state

→ exception\\_review

→ revision\\_pending\\_approval

The reopening action must:

Create a new Master Sales Record version.

Preserve the previous version.

Record a reason.

Record the authorized user.

Identify affected documents.

Identify whether prior signatures are invalidated for the revised scope.

Restart affected approval and signature tasks.

## 6. Transition Guards

Every state transition must evaluate:

Current state

Requested target state

User identity

User role

Record scope

Required permissions

Record version

Required fields

Required approvals

Required signatures

Required payment conditions

Required task completion

Waiting-period calculation

Exception status

Cancellation status

Scope-change status

Audit-event availability

A transition is rejected if any guard fails.

## 7. Master Sales Record Workflow

### 7.1 Create

A Sales Associate creates a draft Master Sales Record for an assigned project.

Required:

Customer

Project

Project location

Project code

Sales Associate

Scope

Payment option

Applicable equipment

Proposed retail price or pricing-pending status

Applicable cost information

### 7.2 Submit

The Sales Associate submits the record for review.

The system validates:

Required fields

Customer consistency

Project consistency

Project-code consistency

Scope consistency

Equipment requirements

Pricing

Costs

Payment option

Duplicate warnings  
Required explanations  
7.3 Approve

A Sales Manager or Comptroller may approve the submitted version.

Approval:

Locks the version.  
Records the approver.  
Records the approval time.  
Identifies the exact version.  
Creates the approval audit event.  
Enables later document generation.

7.4 Reject or return

An authorized approver may reject or return the record for correction.

The system must require:

Rejection or correction reason  
Approver identity  
Date and time  
Affected version  
Audit event

8. Contract and Invoice Dependencies

8.1 Contract dependency

The contract cannot proceed to GECC signature until:

The customer has signed and dated the contract.  
The customer signature is valid.  
The contract version references the approved Master Sales Record.  
The contract has not expired or been superseded.

8.2 Invoice dependency

The invoice cannot proceed to customer signature until:

The contract has all required signatures.  
A GECC authorized signer has signed and dated the invoice.  
The invoice references the approved Master Sales Record.  
The invoice version is current.  
The invoice has not expired or been superseded.

8.3 Installation dependency

An Installation Manager scheduling task cannot be created until:

Contract has all required signatures.  
Invoice has all required signatures.  
The relevant document versions are current.  
No unresolved scope change blocks installation.  
No cancellation is active.

9. Three-Day Installation Rule

9.1 Rule

Installation may not occur earlier than three days after the customer signs both:

The contract  
The invoice

The waiting period begins at the later of the two customer-signature timestamps.

Conceptually:

text

```
installation_eligible_at =
later(contract_customer_signed_at, invoice_customer_signed_at)
+ 3 calendar days
```

## 9.2 Enforcement

The system must:

- Calculate the earliest permitted installation date.
- Display the date to authorized users.
- Prevent scheduling before eligibility.
- Prevent recording installation start before eligibility.
- Prevent recording installation completion before eligibility.
- Display a warning when a proposed date violates the rule.
- Create an audit event for rejected early-installation attempts.

## 9.3 Timezone

The calculation uses:

text

America/New\\_York

The system must preserve the underlying timestamp and display the business-date interpretation in Eastern time.

## 9.4 Exception process

The initial workflow blocks early installation.

If an authorized exception process is implemented later:

- The project enters exception\\_review.
- The exception requires an authorized approver.
- The exception requires a reason.
- The exception requires a date and time.
- The exception does not alter the original signature timestamps.
- The exception is permanently audited.
- The authorized exception must be explicitly linked to the installation event.
- No ordinary user may bypass the three-day rule.

# 10. Installation Task Workflow

## 10.1 Installation scheduling task

The system creates an Installation Manager task after:

- Contract is fully signed.
- Invoice is fully signed.
- The waiting-period requirements are satisfied or an approved exception exists.

The task includes:

Project code  
Customer  
Project location  
Assigned Sales Associate  
Assigned Technician or Technicians  
Approved scope  
Earliest legal installation date  
Current project status  
Required next action  
Due date  
Warning indicator  
10.2 Scheduling  
The Installation Manager may:

Review the task  
Assign or confirm technician assignment  
Enter the installation date  
Confirm that the date is permitted  
Record scheduling notes  
Reschedule before work begins  
The system must reject an installation date earlier than the permitted date.

10.3 Installation start  
Installation start requires:

Assigned Technician  
Approved project scope  
Approved current Master Sales Record  
Valid installation date  
No blocking scope change  
Waiting period satisfied

10.4 Installation completion  
The Technician may record:

Installation completion  
Equipment serial numbers  
Actual materials used  
Completion notes  
Proposed scope changes  
The Technician may not:

Change retail price  
Change approved costs  
Change contract terms  
Approve the project  
Bypass required scope-change review

11. Scope-Change Workflow

11.1 Proposal

If a Technician identifies a scope change:

Technician records the proposed change.  
Technician enters notes and supporting details.  
The project enters scope\_change\_pending.

Work on the changed scope is blocked.  
An approval task is assigned to the Sales Manager or Comptroller.

#### 11.2 Review

The authorized reviewer evaluates:

Proposed scope

Reason

Equipment changes

Serial-number implications

Retail-price changes

Cost changes

Contract impact

Invoice impact

Certificate impact

Installation impact

#### 11.3 Approval

If approved:

The scope change becomes authorized for revision.

Revised price and costs are entered or approved.

A new Master Sales Record version is created.

The original version remains preserved.

Revised documents are generated later.

A new approval and signature sequence is required.

Work cannot continue until required approvals and signatures are complete.

#### 11.4 Rejection

If rejected:

The rejection reason is recorded.

The original approved scope remains controlling.

Work may resume only within the approved scope.

The rejected proposal remains permanently audited.

#### 11.5 Scope-change version rules

A scope change must not:

Edit the original approved version.

Delete original documents.

Remove original signatures.

Change the original project code.

Treat the proposed change as approved before authorization.

### 12. Reopen and Revision Workflow

#### 12.1 Authorized reopening

Reopening is permitted only to authorized users.

The reopening request must include:

Project

Current Master Sales Record version

Reason

Requested changes

Requesting user

Date and time

Approval or authorization